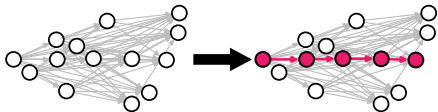
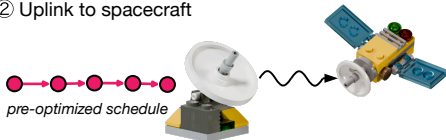


Conventional

① Optimize schedule on the ground



② Uplink to spacecraft



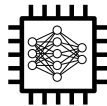
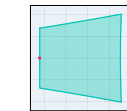
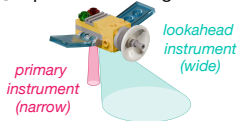
③ Execute schedule exactly



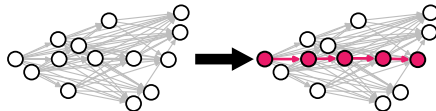
Schedule cannot adapt to changing conditions and can miss imaging utility

Dynamically Tasked (This Work)

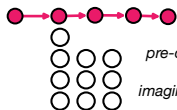
① Spacecraft configuration



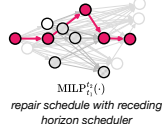
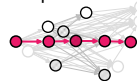
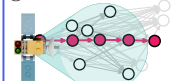
① Optimize schedule on the ground



② Uplink to spacecraft



③ Perform lookaheads and repair schedule



Dynamic tasking with onboard inference repair the schedule, mitigating low-utility occluded tasks and incorporating alternates